

Nav-si

Privacy Policy

Last updated: May 3, 2026

1. Overview

Nav-si is an open-source assistive navigation application designed to empower blind and low-vision users through AI-assisted object detection, scene description, and spatial awareness. We are committed to processing your data as minimally and transparently as possible.

Nav-si operates in two distinct modes, and understanding them is central to understanding how your data is handled:

- **Local Mode (default):** Object detection runs entirely on your device using an on-device YOLOv11n model. Video frames are only transmitted over the Internet if the wearable camera is being used. Only anonymised bounding box coordinates and class labels are used locally.
- **Cloud Mode (opt-in):** When you choose to use cloud-based scene description, image frames are encrypted and sent to a processing server. This mode provides richer descriptions but involves data leaving your device (if used with the wearable camera, video is sent directly from the camera to the server which sends anonymous bounding box data back to the phone).

Local Mode is the default. Cloud Mode requires an explicit opt-in and can be toggled off at any time from within the app settings.

2. Data We Collect

a. Information You Provide

- **Feedback or Support Messages:** Text or audio content you voluntarily submit when contacting us for support.
- **Camera-Device Pairing Data:** A locally generated pairing identifier used to associate your phone with a Nav-si-compatible camera accessory. This is stored locally and only transmitted to us as a way of authenticating your device.

b. Automatically Collected Information (Local Mode)

In Local Mode, no personal data leaves your device. The following is processed entirely on-device:

- Bounding box coordinates and object class labels (e.g. "person", "door", "car") generated by the local YOLOv11n model.
- Crash reports and diagnostic events, which are anonymised before transmission and contain no image or location data.
- App version and OS version for compatibility and update purposes.

Even in Local Mode, bounding box data combined with a consistent device identifier could theoretically reveal patterns about your routine. We mitigate this by never pairing bounding box data with persistent user identifiers on our systems.

c. Automatically Collected Information (Cloud Mode, opt-in only)

When Cloud Mode is enabled, the following is transmitted to our secure processing server:

- Encrypted image frames from your device camera, processed solely to generate scene descriptions.
- Session tokens: Short-lived tokens used to route results to the correct device. These expire automatically and are not retained.

We do not collect or store:

- Precise or approximate location data.
- Audio or speech data.
- Persistent device identifiers or advertising IDs.
- Any data for advertising or profiling purposes.

3. How We Use Your Data

We use data only to:

- Deliver core navigation and object detection functionality.
- Generate scene descriptions (Cloud Mode only, when opted in).
- Diagnose and resolve technical issues via anonymised crash reports.
- Notify you of important security or software updates.
- Comply with applicable laws and accessibility regulations.

We do not use your data for advertising. We do not sell your data to third parties.

Model Improvement (Opt-In Only)

We offer a separate, optional programme to help improve Nav-si's detection accuracy. If you choose to participate:

- All direct personal identifiers are removed before any data is processed.
- You can withdraw from this programme at any time from app settings, and all your contributed data will be deleted.

This opt-in is separate from general app usage. You will not be enrolled automatically.

4. Camera Hardware and Device Pairing

If you use a Nav-si-compatible camera accessory, the following applies:

- Pairing is performed via QR code scan to establish a shared cryptographic secret between your phone and camera. This makes it significantly harder for a third party to intercept or spoof the pairing.

- Pairing credentials are stored in encrypted non-volatile storage on the camera hardware, and in your device's secure enclave (Android Keystore / iOS Secure Enclave) on the phone.
- In Cloud Mode, the camera streams encrypted video to the processing server, which routes results back to your specific paired device only. The server does not retain this stream after processing.
- In Local Mode, no data from the camera is transmitted to any server. All processing occurs on your device.

5. Data Sharing

We share limited data only with trusted service providers who support our operations:

- Cloud Infrastructure (e.g. processing servers) for secure, temporary image processing in Cloud Mode.
- Analytics Services for aggregated and fully anonymised performance metrics — no individual user data is shared.
- Support Platforms when you contact us directly for help.

All third-party partners are bound by strict data protection agreements consistent with UK GDPR and applicable privacy law. We do not share data with advertising networks, data brokers, or social media platforms.

6. Data Retention

- Cloud Mode image frames are deleted from our servers immediately after processing. Unless the user has chosen to, data is never stored persistently.
- Session tokens expire automatically and are purged within 24 hours.
- Anonymised crash logs and diagnostics are retained for up to 30 days.
- Support messages are retained only for as long as necessary to resolve your issue and deleted thereafter.
- If you opt in to the model improvement programme, anonymised bounding box data is retained until you withdraw consent, at which point it is deleted.

You can request complete deletion of all data associated with your use of Nav-si at any time. See Section 8 for how to do this.

7. Security

We implement the following technical and organisational safeguards:

In Transit

- All data transmitted to our servers uses TLS encryption. The app implements certificate pinning to ensure it only communicates with servers presenting the correct certificate, preventing man-in-the-middle attacks.

- A certificate pin rotation plan is maintained so that a certificate renewal does not disrupt service. Backup pins are in place at all times.

On Device

- Pairing credentials and session tokens are stored in your device's hardware-backed secure storage (Android Keystore / iOS Secure Enclave).
- The app is built with dependency integrity checks and uses a locked dependency manifest to prevent supply chain attacks.
- App logs are retained on-device for a maximum of 7 days and contain no image or location data.

On Our Servers

- Image frames are processed in an isolated inference environment, separate from user account infrastructure, so a breach of one cannot expose the other.
- Input validation (image dimensions and file type verification) is applied before any data reaches the processing model.
- Access to production systems is restricted to authorised personnel only, with audit logging of all access events.
- Device attestation is used (Android Play Integrity API; Apple DeviceCheck) to verify that requests originate from unmodified, legitimate installations of Nav-si.
- Rate limiting and exponential back-off are enforced on all credential endpoints to prevent brute-force attacks.

Camera Firmware

- Nav-si-compatible camera firmware uses hardware specific tools to secure your data on-device.
- Over-the-air firmware updates are signed and verified before installation.

8. Your Rights

You have the following rights regarding your personal data under UK GDPR and applicable law:

- Access: Request a copy of the personal data we hold about you.
- Correction: Ask us to correct inaccurate or incomplete data.
- Deletion: Request that we delete your data. We will action this within 30 days.
- Portability: Receive your data in a structured, machine-readable format.
- Withdraw Consent: Opt out of Cloud Mode or the model improvement programme at any time via app settings, without affecting your ability to use Local Mode.

To exercise any of these rights, contact us at: privacy@nav-si.app

We will verify your identity before taking action on any request.

9. Children's Privacy

Nav-si is not directed at individuals under 18 years of age. We do not knowingly collect personal data from minors. If you believe a minor has submitted data to us, please contact us at privacy@nav-si.app and we will delete it promptly.

10. UK GDPR and Accessibility Considerations

Nav-si processes visual scene data that may incidentally capture information about third parties (e.g. people in a user's environment). We have considered this carefully:

- In Local Mode, no such data ever leaves the device, and bounding box labels are not stored persistently.
- In Cloud Mode, frames are processed and then deleted immediately (unless users opt-into model training services).

As an assistive technology application that processes visual data of users' environments, we have conducted a Data Protection Impact Assessment (DPIA) to identify and mitigate privacy risks. A summary is available on request.

Bounding box metadata (e.g. labels such as "person" or "car") may, in combination with a user ID and timestamp, constitute personal data under UK GDPR where it reveals behavioural patterns. We mitigate this by never linking bounding box data to persistent user identifiers on our systems.

11. Changes to This Policy

We may update this policy to reflect changes in Nav-si's features or applicable legal requirements. When we make material changes, we will notify you via an in-app notification prior to the change taking effect.

The latest version of this policy is always available in the GitHub page: <https://github.com/ComsKeaneDev/nav-si>, and within the Nav-si app.

Nav-si is open-source software. This policy applies to the official Nav-si application only.